



Analysis and Design of Algorithms



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رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي



O my Lord! Expand for me my chest [with assurance] and ease for me my task and untie the knot from my tongue that they may understand my speech. (Quran 20 : 25-28)



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Algorithm Analysis

Recurrence
(Formal Analysis)



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Introduction

- Iterative Algorithms
 - Informal Analysis
 - Formal Analysis
 - RAM Model Analysis
- **Recursive Algorithm**
 - Informal Analysis
 - Formal Analysis
 - Backwards Substitution Method
 - Substitution Method
 - **Recursion Tree Method**
 - Master Theorem Method

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Merge Sort

```
MERGE-SORT( $A, p, r$ )  
1  if  $p < r$   
2    then  $q \leftarrow \lfloor (p + r)/2 \rfloor$   
3        MERGE-SORT( $A, p, q$ )  
4        MERGE-SORT( $A, q + 1, r$ )  
5        MERGE( $A, p, q, r$ )
```

$$\begin{aligned} - T(n) &= \Theta(1) && \text{if } n = 1 \\ &= 2 T(n/2) + \Theta(n) && \text{if } n > 1 \\ &= 2 T(n/2) + c n \end{aligned}$$

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Merge Sort Recurrence

$$\begin{aligned} - T(n) &= \Theta(1) && \text{if } n = 1 \\ &= 2 T(n/2) + \Theta(n) && \text{if } n > 1 \\ &= 2 T(n/2) + c n \end{aligned}$$

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Steps for Recursion Tree Method

- Construct the Tree
- Find the problem size on each level
- Determine the cost for each node
- Find no. of nodes on each Level
- Determine the cost of each level
- Determine the total no. of levels
- Determine the last level and its cost
- Sum the cost of each level
- Verify the results with help of Substitution Method

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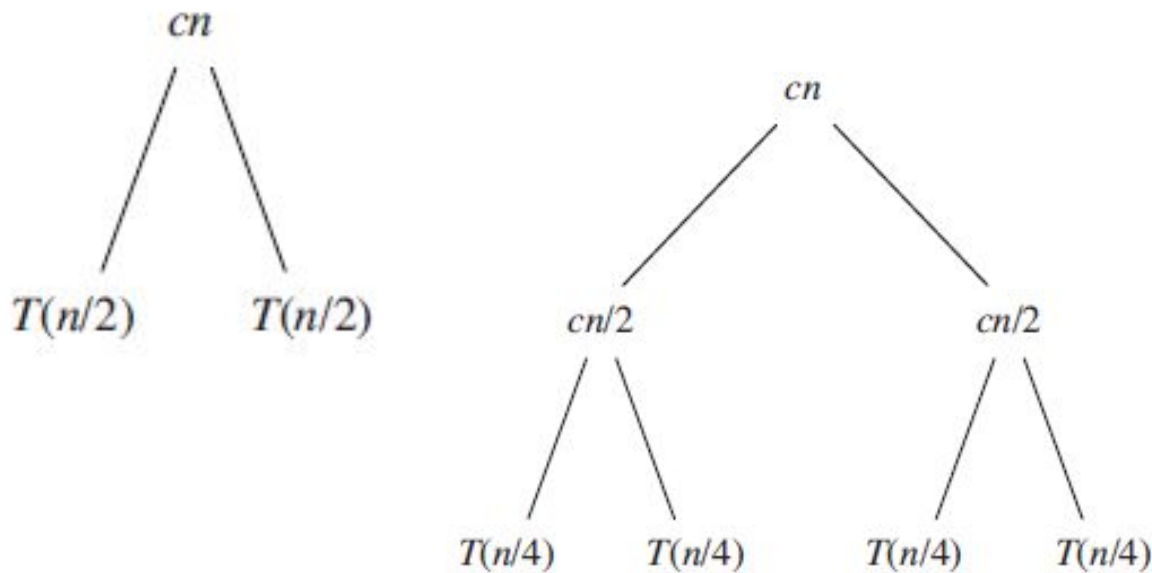


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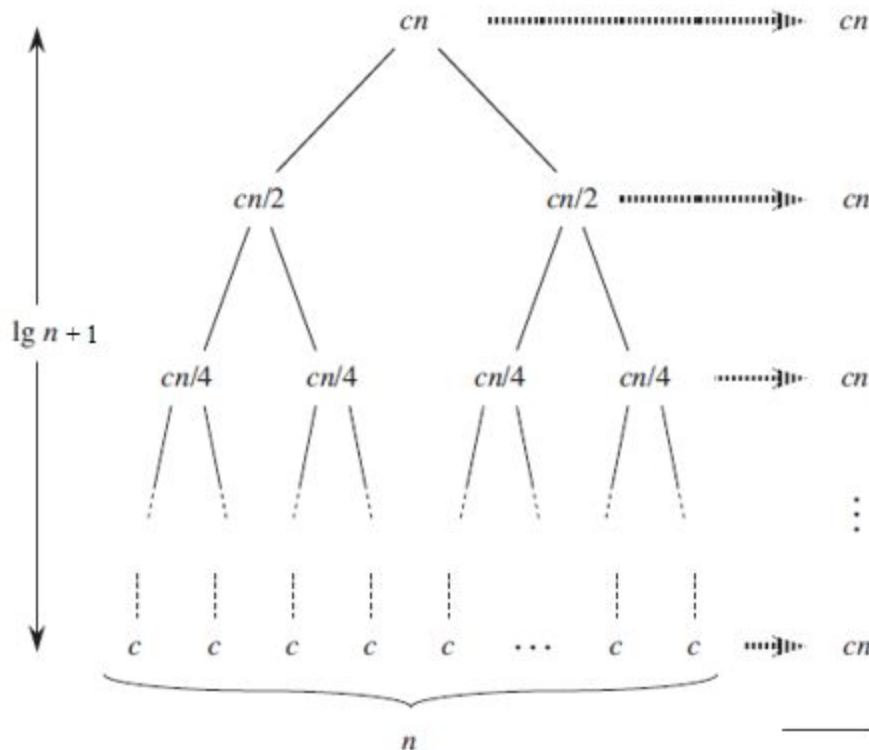
Recursion Tree - $T(n) = 2T(n/2) + cn$

$T(n)$



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Recursion Tree ($T(n) = 2T(n/2) + cn$)



$$\begin{aligned}
 \text{Total} &= \text{Cost at each level} * \text{No. of levels} \\
 &= cn (\lg n + 1) \\
 &= cn \lg n + cn
 \end{aligned}$$

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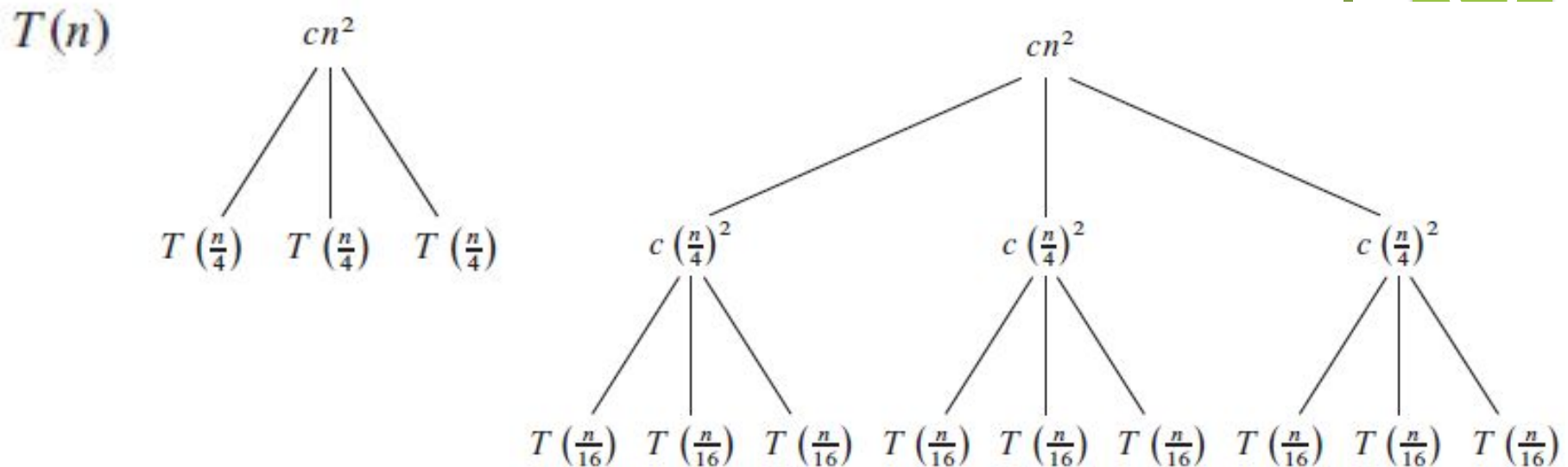


Recursion Tree Method

- $T(n) = 3 T (\text{floor}(n/4)) + \Theta n^2$

Recursion Tree – $T(n) = 3 T(\text{floor}(n/4)) + c n^2$

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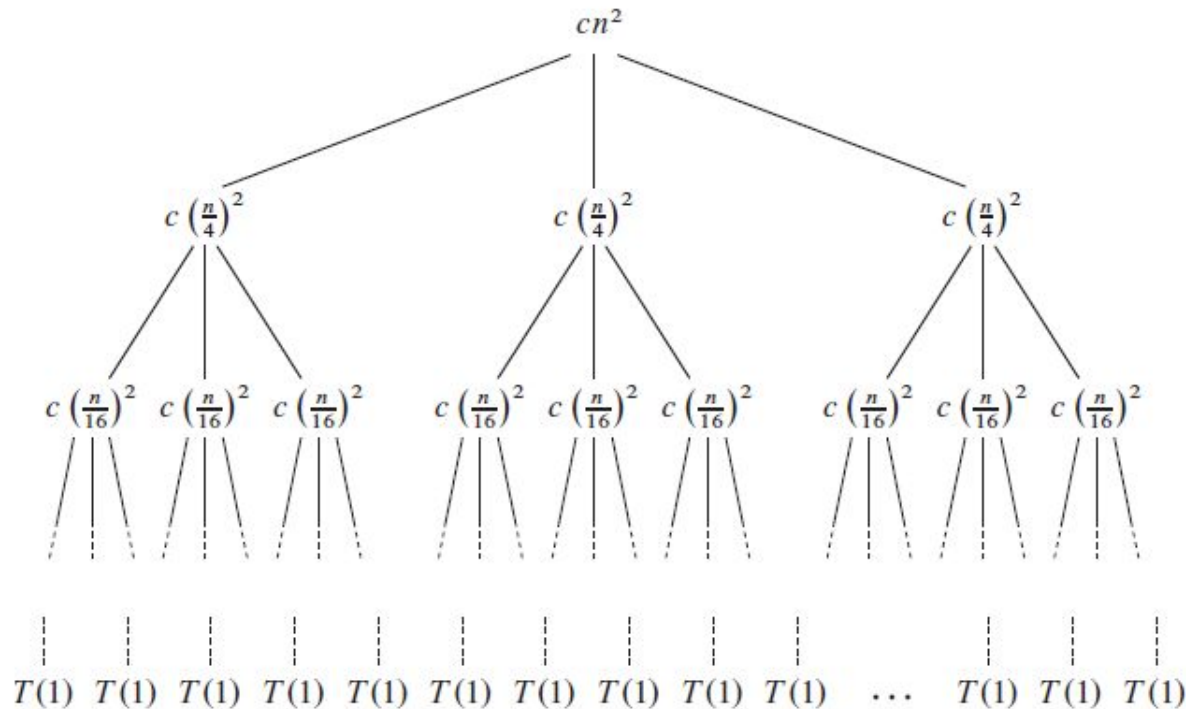




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Recursion Tree – $T(n) = 3 T(\text{floor}(n/4)) + c n^2$



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Recursion Tree – $T(n) = 3 T(\text{floor}(n/4)) + c n^2$

- Problem Size at level “i” $n/4^i$
- Cost of each node at level “i” $c n^2/16^i$
- No. of nodes at level “i” 3^i
- Total cost at level “i” $(3/16)^i c n^2$
- Last Level number $\log_4 n$
- No. of nodes at last Level $n^{\log_4 3}$
- Cost at last Level $\Theta(n^{\log_4 3})$

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Recursion Tree – $T(n) = 3 T(\text{floor}(n/4)) + c n^2$

Total cost at level 0 + level 1 + level 2 ++ second last level + Last level

$$T(n) = cn^2 + \frac{3}{16}cn^2 + \left(\frac{3}{16}\right)^2 cn^2 + \dots + \left(\frac{3}{16}\right)^{\log_4 n - 1} cn^2 + \Theta(n^{\log_4 3})$$

$$= \sum_{i=0}^{\log_4 n - 1} \left(\frac{3}{16}\right)^i cn^2 + \Theta(n^{\log_4 3})$$

$$< \sum_{i=0}^{\infty} \left(\frac{3}{16}\right)^i cn^2 + \Theta(n^{\log_4 3})$$

$$< \frac{1}{1 - (3/16)} cn^2 + \Theta(n^{\log_4 3})$$

$$< \frac{16}{13} cn^2 + \Theta(n^{\log_4 3})$$

$$= O(n^2)$$

Infinite geometric series, where
first term, $a = (3/16)^0 = 1$
common ratio, $r = 3/16$
So, sum $S = a / (1 - r)$

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Recursion Tree – $T(n) = 3T(\text{floor}(n/4)) + cn^2$

- Verification by Substitution Method

$$\begin{aligned}T(n) &\leq 3T(\lfloor n/4 \rfloor) + cn^2 \\&\leq 3d \lfloor n/4 \rfloor^2 + cn^2 \\&\leq 3d(n/4)^2 + cn^2 \\&= \frac{3}{16}dn^2 + cn^2 \\&\leq dn^2,\end{aligned}$$

Cost of problem size 'n'

Cost of problem size 'n/4' where 'd' is any constant

We have proved by using recursion tree that $T(n) = cn^2$

$$d \geq (16/13)c$$

For example, if we take $d = (16/13)c$, then above expression can be solved as $(3/16)dn^2 + cn^2 = (3/16)(16/13)cn^2 + cn^2 = 3/13cn^2 + cn^2 = (16/13)cn^2$

Which is similar to the one proved by recursion tree

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Recursion Tree

- $T(n) = T(n/3) + T(2n/3) + O(n)$

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(Since it is binary tree)

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$$T(n) = T(n/3) + T(2n/3) + cn$$

$$T(n) = T(n/3) + T(2n/3) + cn$$

$$\leq d(n/3) \lg(n/3) + d(2n/3) \lg(2n/3) + cn$$

$$= d(n/3)[\lg n - \lg 3] + d(2n/3)[\lg n - \lg(3/2)] + cn$$

$$= d(n/3) \lg n - d(n/3) \lg 3 + d(2n/3) \lg n - d(2n/3) \lg(3/2) + cn$$

$$= d(n/3) \lg n + d(2n/3) \lg n - d(n/3) \lg 3 - d(2n/3)[\lg 3 - \lg 2] + cn$$

$$= d(n/3 + 2n/3) \lg n - d(n/3) \lg 3 - d(2n/3)[\lg 3 - 1] + cn$$

$$= d n \lg n - d(n/3) \lg 3 - d(2n/3) \lg 3 + d(2n/3) + cn$$

$$= d n \lg n - d[n/3 + 2n/3] \lg 3 + d(2n/3) + cn$$

$$= d n \lg n - d n \lg 3 + d(2n/3) + cn$$

$$= d n \lg n - d n (\lg 3 - 2/3) + cn$$

$$\leq d n \lg n$$

This inequality holds as long as $d \geq c/(\lg 3 - 2/3)$

(because $d n \lg n - d n (\lg 3 - 2/3) + cn$ will become

$$d n \lg n - c/(\lg 3 - 2/3) n (\lg 3 - 2/3) + cn$$

$$= d n \lg n - cn + cn$$

We have proved by using recursion tree that

$$T(n) = O(n \lg n)$$

$$\boxed{\log(2n/3) = \log(n/3/2)}$$

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Allah (Alone) is Sufficient for us, and He is the Best Disposer of affairs (for us). (Quran 3 : 173)